

Linux 命令实践

一、目录查看命令 (ls)

```
tisoft@Tisoft:~/linux-command-practice$ cd ./01-navigation/
tisoft@Tisoft:~/linux-command-practice/01-navigation$ ls
docs empty-dir images long-file-name-for-ls-practice.txt meeting-link readme.txt
tisoft@Tisoft:~/linux-command-practice/01-navigation$ ls -a
. .hidden-note empty-dir long-file-name-for-ls-practice.txt readme.txt
.. docs images meeting-link
tisoft@Tisoft:~/linux-command-practice/01-navigation$ ls -l
total 20
drwxr-xr-x 3 tisoft tisoft 4096 Sep 21 08:15 docs
drwxr-xr-x 2 tisoft tisoft 4096 Sep 21 08:15 empty-dir
drwxr-xr-x 2 tisoft tisoft 4096 Sep 21 08:15 images
-rw-r--r-- 1 tisoft tisoft 40 Sep 21 08:15 long-file-name-for-ls-practice.txt
lrwxrwxrwx 1 tisoft tisoft 15 Sep 21 08:15 meeting-link -> docs/meeting.md
-rw-r--r-- 1 tisoft tisoft 35 Sep 21 08:15 readme.txt
tisoft@Tisoft:~/linux-command-practice/01-navigation$ ls -lh
total 20K
drwxr-xr-x 3 tisoft tisoft 4.0K Sep 21 08:15 docs
drwxr-xr-x 2 tisoft tisoft 4.0K Sep 21 08:15 empty-dir
drwxr-xr-x 2 tisoft tisoft 4.0K Sep 21 08:15 images
-rw-r--r-- 1 tisoft tisoft 40 Sep 21 08:15 long-file-name-for-ls-practice.txt
lrwxrwxrwx 1 tisoft tisoft 15 Sep 21 08:15 meeting-link -> docs/meeting.md
-rw-r--r-- 1 tisoft tisoft 35 Sep 21 08:15 readme.txt
tisoft@Tisoft:~/linux-command-practice/01-navigation$ ls -lt
total 20
lrwxrwxrwx 1 tisoft tisoft 15 Sep 21 08:15 meeting-link -> docs/meeting.md
drwxr-xr-x 3 tisoft tisoft 4096 Sep 21 08:15 docs
drwxr-xr-x 2 tisoft tisoft 4096 Sep 21 08:15 images
-rw-r--r-- 1 tisoft tisoft 40 Sep 21 08:15 long-file-name-for-ls-practice.txt
-rw-r--r-- 1 tisoft tisoft 35 Sep 21 08:15 readme.txt
drwxr-xr-x 2 tisoft tisoft 4096 Sep 21 08:15 empty-dir
tisoft@Tisoft:~/linux-command-practice/01-navigation$ ls ~/linux-command-practice/02-search/
large-sample.bin logs notes test-result.txt
```

二、用户与路径查看命令 (who、pwd)

```
tisoft@Tisoft:~/linux-command-practice/01-navigation$ who
tisoft pts/1 2026-09-21 08:00
tisoft@Tisoft:~/linux-command-practice/01-navigation$ whoami
tisoft
tisoft@Tisoft:~/linux-command-practice/01-navigation$ who -u
tisoft pts/1 2026-09-21 08:00 00:30 453
tisoft@Tisoft:~/linux-command-practice/01-navigation$ who -q
tisoft
# users=1
tisoft@Tisoft:~/linux-command-practice/01-navigation$ who -b
system boot 2026-09-21 08:00
tisoft@Tisoft:~/linux-command-practice/01-navigation$ pwd
/home/tisoft/linux-command-practice/01-navigation
tisoft@Tisoft:~/linux-command-practice/01-navigation$ pwd -P
/home/tisoft/linux-command-practice/01-navigation
```

三、目录切换命令 (cd)

```
tisoft@Tisoft:~/linux-command-practice$ cd ~/linux-command-practice/
tisoft@Tisoft:~/linux-command-practice$ pwd
/home/tisoft/linux-command-practice
tisoft@Tisoft:~/linux-command-practice$ cd 01-navigation/
tisoft@Tisoft:~/linux-command-practice/01-navigation$ pwd
/home/tisoft/linux-command-practice/01-navigation
tisoft@Tisoft:~/linux-command-practice/01-navigation$ cd docs
tisoft@Tisoft:~/linux-command-practice/01-navigation/docs$ pwd
/home/tisoft/linux-command-practice/01-navigation/docs
tisoft@Tisoft:~/linux-command-practice/01-navigation/docs$ cd ..
tisoft@Tisoft:~/linux-command-practice/01-navigation$ pwd
/home/tisoft/linux-command-practice/01-navigation
tisoft@Tisoft:~/linux-command-practice/01-navigation$ cd -
/home/tisoft/linux-command-practice/01-navigation/docs
tisoft@Tisoft:~/linux-command-practice/01-navigation/docs$ pwd
/home/tisoft/linux-command-practice/01-navigation/docs
tisoft@Tisoft:~/linux-command-practice/01-navigation/docs$ cd
tisoft@Tisoft:~$ pwd
/home/tisoft
tisoft@Tisoft:~$ cd /
tisoft@Tisoft:/$ pwd
/
tisoft@Tisoft:/$ cd ~/linux-command-practice/
tisoft@Tisoft:~/linux-command-practice$ |
```

四、帮助文档命令 (man)

```
tisoft@Tisoft: ~/linux-commma x + v
LS(1) User Commands LS(1)
NAME
ls - list directory contents
SYNOPSIS
ls [OPTION]... [FILE]...
DESCRIPTION
List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.

-a, --all
    do not ignore entries starting with .

-A, --almost-all
    do not list implied . and ..

--author
    with -l, print the author of each file

-b, --escape
    print C-style escapes for nongraphic characters

--block-size=SIZE
    with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below

-B, --ignore-backups
    do not list implied entries ending with ~

-c    with -lt: sort by, and show, ctime (time of last change of file status information);
      with -l: show ctime and sort by name; otherwise: sort by ctime, newest first

-C    list entries by columns

--color[=WHEN]
    color the output WHEN; more info below

-d, --directory
    list directories themselves, not their contents

-D, --dired
    generate output designed for Emacs' dired mode

-f    list all entries in directory order

-F, --classify[=WHEN]
Manual page ls(1) line 1 (press h for help or q to quit)
```

五、命令定位 (whereis、which、type)

```
tisoft@Tisoft:~/linux-command-practice$ whereis ls
ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz
tisoft@Tisoft:~/linux-command-practice$ whereis -b ls
ls: /usr/bin/ls
tisoft@Tisoft:~/linux-command-practice$ whereis -m ls
ls: /usr/share/man/man1/ls.1.gz
tisoft@Tisoft:~/linux-command-practice$ whereis find
find: /usr/bin/find /usr/share/man/man1/find.1.gz /usr/share/info/find.info-2.gz /usr/share/info/find.info.gz /usr/share/info/find.info-1.gz
tisoft@Tisoft:~/linux-command-practice$ which ls
/usr/bin/ls
tisoft@Tisoft:~/linux-command-practice$ which cd
tisoft@Tisoft:~/linux-command-practice$ which grep
/usr/bin/grep
tisoft@Tisoft:~/linux-command-practice$ which -a python python3
/usr/bin/python3
/bin/python3
tisoft@Tisoft:~/linux-command-practice$ type cd
cd is a shell builtin
```

`which cd` 没有输出，因为 `cd` 是 Shell 内建命令，并非独立可执行文件；`type cd` 可以正确识别。

六、文件查找命令 (find)

```
tisoft@Tisoft:~/linux-command-practice$ find 02-search -name '*test*'
02-search/test-result.txt
tisoft@Tisoft:~/linux-command-practice$ find 02-search -name '*.md'
02-search/notes/homework.md
tisoft@Tisoft:~/linux-command-practice$ find 02-search -size +100k -ls
13767 128 -rw-r--r-- 1 tisoft tisoft 131072 Sep 21 08:15 02-search/large-sample.bin
tisoft@Tisoft:~/linux-command-practice$ find 04-permissions/ -perm 600 -ls
13812 4 -rw----- 1 tisoft tisoft 44 Sep 21 08:15 04-permissions/private/secret.txt
tisoft@Tisoft:~/linux-command-practice$ find 02-search/ -type f
02-search/logs/old/app-2025.log
02-search/logs/app.log
02-search/large-sample.bin
02-search/notes/linux-notes.txt
02-search/notes/homework.md
02-search/test-result.txt
tisoft@Tisoft:~/linux-command-practice$ find . -type d -empty
./work
./01-navigation/empty-dir
./screenshots
tisoft@Tisoft:~/linux-command-practice$ |
```

七、索引查找命令 (locate)

```
tisoft@Tisoft:~/linux-command-practice$ locate passwd | head
/etc/passwd
/etc/passwd-
/etc/pam.d/chpasswd
/etc/pam.d/passwd
/etc/security/opasswd
/mnt/c/Program Files/Git/usr/bin/mkpasswd.exe
/mnt/c/Program Files/Git/usr/bin/passwd.exe
/mnt/c/Program Files/Git/usr/share/awk/ns_passwd.awk
/mnt/c/Program Files/Git/usr/share/awk/passwd.awk
/mnt/c/Program Files/Git/usr/share/vim/vim92/ftplugin/passwd.vim
tisoft@Tisoft:~/linux-command-practice$ locate -i README.md | head
/home/tisoft/linux-command-practice/README.md
/home/tisoft/linux-command-practice/06-archive/project/README.md
/mnt/c/Program Files/Git/mingw64/doc/git-credential-manager/README.md
/mnt/c/Program Files/Git/mingw64/share/doc/git-lfs/README.md
/mnt/c/Program Files/Git/usr/share/mintty/sounds/README.md
/mnt/c/Program Files/Git/usr/share/vim/vim92/pack/dist/opt/editorconfig/README.md
/mnt/c/Program Files/Git/usr/share/vim/vim92/pack/dist/opt/netrw/README.md
/mnt/c/Program Files/WindowsPowerShell/Modules/Pester/3.4.0/README.md
/mnt/c/Program Files/nodejs/node_modules/corepack/README.md
/mnt/c/Program Files/nodejs/node_modules/npm/README.md
tisoft@Tisoft:~/linux-command-practice$ locate linux-command-practice | head -n 20
/home/tisoft/linux-command-practice
/home/tisoft/linux-command-practice/01-navigation
/home/tisoft/linux-command-practice/02-search
/home/tisoft/linux-command-practice/03-sort
/home/tisoft/linux-command-practice/04-permissions
/home/tisoft/linux-command-practice/05-binary
/home/tisoft/linux-command-practice/06-archive
/home/tisoft/linux-command-practice/PRACTICE-CHECKLIST.md
/home/tisoft/linux-command-practice/README.md
/home/tisoft/linux-command-practice/01-navigation/.hidden-note
/home/tisoft/linux-command-practice/01-navigation/docs
/home/tisoft/linux-command-practice/01-navigation/empty-dir
/home/tisoft/linux-command-practice/01-navigation/images
/home/tisoft/linux-command-practice/01-navigation/long-file-name-for-ls-practice.txt
/home/tisoft/linux-command-practice/01-navigation/meeting-link
/home/tisoft/linux-command-practice/01-navigation/readme.txt
/home/tisoft/linux-command-practice/01-navigation/docs/archive
/home/tisoft/linux-command-practice/01-navigation/docs/meeting.md
/home/tisoft/linux-command-practice/01-navigation/docs/archive/report-old.txt
/home/tisoft/linux-command-practice/01-navigation/images/photo.txt
tisoft@Tisoft:~/linux-command-practice$ locate '*.conf' | head
/etc/adduser.conf
/etc/ca-certificates.conf
/etc/debconf.conf
/etc/deluser.conf
/etc/dhclient.conf
/etc/e2scrub.conf
/etc/fuse.conf
/etc/gai.conf
/etc/host.conf
/etc/ld.so.conf
tisoft@Tisoft:~/linux-command-practice$ locate -l 10 '.txt'
/etc/X11/rgb.txt
/home/tisoft/linux-command-practice/01-navigation/long-file-name-for-ls-practice.txt
/home/tisoft/linux-command-practice/01-navigation/readme.txt
/home/tisoft/linux-command-practice/01-navigation/docs/archive/report-old.txt
/home/tisoft/linux-command-practice/01-navigation/images/photo.txt
/home/tisoft/linux-command-practice/02-search/test-result.txt
/home/tisoft/linux-command-practice/02-search/notes/linux-notes.txt
/home/tisoft/linux-command-practice/03-sort/ip-addresses.txt
/home/tisoft/linux-command-practice/03-sort/numbers.txt
/home/tisoft/linux-command-practice/03-sort/words.txt
tisoft@Tisoft:~/linux-command-practice$ |
```

八、文本搜索命令 (grep)


```

tisoft@Tisoft:~/linux-command-practice$ grep ERROR 02-search/logs/app.log
ERROR database unavailable
tisoft@Tisoft:~/linux-command-practice$ grep -i error 02-search/notes/linux-notes.txt
Error and ERROR differ by case.
tisoft@Tisoft:~/linux-command-practice$ grep -n ERROR 02-search/logs/app.log
2:ERROR database unavailable
tisoft@Tisoft:~/linux-command-practice$ grep -R -n ERROR 02-search
02-search/logs/old/app-2025.log:2:ERROR old timeout
02-search/logs/app.log:2:ERROR database unavailable
02-search/notes/linux-notes.txt:3:Error and ERROR differ by case.
tisoft@Tisoft:~/linux-command-practice$ grep -R -n 'error' 02-search
tisoft@Tisoft:~/linux-command-practice$ grep -R -n -i 'error' 02-search
02-search/logs/old/app-2025.log:2:ERROR old timeout
02-search/logs/app.log:2:ERROR database unavailable
02-search/notes/linux-notes.txt:3:Error and ERROR differ by case.
tisoft@Tisoft:~/linux-command-practice$ grep -v root /etc/passwd | head
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
tisoft@Tisoft:~/linux-command-practice$ |

```

`grep -R -n 'error'` 没有匹配结果，是因为 `grep` 默认区分大小写；增加 `-i` 后可以匹配。

九、输出与重定向命令（echo）

```

tisoft@Tisoft:~/linux-command-practice$ echo 'Hello world'
Hello world
tisoft@Tisoft:~/linux-command-practice$ echo 'new content' > work/test.txt
tisoft@Tisoft:~/linux-command-practice$ cat work/test.txt
new content
tisoft@Tisoft:~/linux-command-practice$ echo 'add one line' >> work/test.txt
tisoft@Tisoft:~/linux-command-practice$ cat work/test.txt
new content
add one line
tisoft@Tisoft:~/linux-command-practice$ echo "$PATH"
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/w
sl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.12.0_x6
4_8wekyb3d8bbwe:/mnt/c/Python314/Scripts:/mnt/c/Python314:/mnt/c/Program Files (x86)/VMware/VMwa
re Workstation/bin:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Win
dows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NV
IDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/WINDOWS
/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.
0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/nodejs:/mnt/c/ProgramData/chocolatey/bin:
/mnt/c/Program Files/Git/cmd:/mnt/c/Users/12441/.kimi-code/bin:/mnt/c/Users/12441/AppData/Local/Mic
rosoft/WindowsApps:/mnt/c/Users/12441/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/124
41/AppData/Roaming/npm:/snap/bin
tisoft@Tisoft:~/linux-command-practice$ echo "$USER"
tisoft
tisoft@Tisoft:~/linux-command-practice$ |

```

十、二进制内容查看命令（od）

```
tisoft@Tisoft:~/linux-command-practice$ cd 05-binary/
tisoft@Tisoft:~/linux-command-practice/05-binary$ od test.bin
0000000 042577 043114 000402 000001 071160 061541 064564 062543
0000020 061055 067151 071141 000171 001001 000377
0000033
tisoft@Tisoft:~/linux-command-practice/05-binary$ od -Ax -tx1z test.bin
000000 7f 45 4c 46 02 01 01 00 70 72 61 63 74 69 63 65 >.ELF....practice<
000010 2d 62 69 6e 61 72 79 00 01 02 ff >-binary....<
00001b
tisoft@Tisoft:~/linux-command-practice/05-binary$ od -c test.txt
0000000 H e l l o L i n u x ! \n A S C
0000020 I I a n d b i n a r y i n
0000040 s p e c t i o n . \n
0000052
tisoft@Tisoft:~/linux-command-practice/05-binary$ od -N 8 -tx1 test.bin
0000000 7f 45 4c 46 02 01 01 00
0000010
tisoft@Tisoft:~/linux-command-practice/05-binary$
```

十一、文本排序命令 (sort)

```
tisoft@Tisoft:~/linux-command-practice/03-sort$ sort words.txt
Orange
apple
apple
banana
grape
pear
tisoft@Tisoft:~/linux-command-practice/03-sort$ sort -f words.txt
apple
apple
banana
grape
Orange
pear
tisoft@Tisoft:~/linux-command-practice/03-sort$ sort -n numbers.txt
2
2
3
10
21
100
tisoft@Tisoft:~/linux-command-practice/03-sort$ sort -nr numbers.txt
100
21
10
3
2
2
tisoft@Tisoft:~/linux-command-practice/03-sort$ sort -nu numbers.txt
2
3
10
21
100
tisoft@Tisoft:~/linux-command-practice/03-sort$ sort -V ip-addresses.txt
10.0.0.2
10.0.0.10
172.16.0.1
192.168.1.2
192.168.1.10
tisoft@Tisoft:~/linux-command-practice/03-sort$ sort -t, -k2,2n scores.csv
bob,72
dave,72
alice,88
carol,95
tisoft@Tisoft:~/linux-command-practice/03-sort$ |
```

十二、文件与目录操作（cp、mv、mkdir、rm、cat、head、tail、more、less）


```

tisoft@Tisoft:~/linux-command-practice/work$ rm -rf file-demo
tisoft@Tisoft:~/linux-command-practice/work$ mkdir -p file-demo/source file-demo/empty-dir
tisoft@Tisoft:~/linux-command-practice/work$ printf 'line 1\nline 2\nline 3\nline 4\nline 5\n' > file-demo/source/sample.txt
tisoft@Tisoft:~/linux-command-practice/work$ cd file-demo
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ pwd
/home/tisoft/linux-command-practice/work/file-demo
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ cp source/sample.txt copy.txt
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ mv copy.txt renamed.txt
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ mkdir new-dir
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ cat renamed.txt
line 1
line 2
line 3
line 4
line 5
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ head -n 2 renamed.txt
line 1
line 2
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ tail -n 2 renamed.txt
line 4
line 5
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ more renamed.txt
line 1
line 2
line 3
line 4
line 5
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ less renamed.txt
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ rm renamed.txt
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ rmdir empty-dir
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ ls -la
total 16
drwxr-xr-x 4 tisoft tisoft 4096 Sep 21 08:49 .
drwxr-xr-x 3 tisoft tisoft 4096 Sep 21 08:48 ..
drwxr-xr-x 2 tisoft tisoft 4096 Sep 21 08:48 new-dir
drwxr-xr-x 2 tisoft tisoft 4096 Sep 21 08:48 source
tisoft@Tisoft:~/linux-command-practice/work/file-demo$ |

```

十三、文件权限管理（chmod、umask、stat）

```

tisoft@Tisoft:~/linux-command-practice/04-permissions$ ls -l
total 12
-rw-r--r-- 1 tisoft tisoft 59 Sep 21 08:15 hello.sh
drwx----- 2 tisoft tisoft 4096 Sep 21 08:15 private
-r--r--r-- 1 tisoft tisoft 17 Sep 21 08:15 read-only.txt
tisoft@Tisoft:~/linux-command-practice/04-permissions$ ./hello.sh
-bash: ./hello.sh: Permission denied
tisoft@Tisoft:~/linux-command-practice/04-permissions$ chmod u+x hello.sh
tisoft@Tisoft:~/linux-command-practice/04-permissions$ ls -l hello.sh
-rwxr--r-- 1 tisoft tisoft 59 Sep 21 08:15 hello.sh
tisoft@Tisoft:~/linux-command-practice/04-permissions$ ./hello.sh
Hello from an executable script
tisoft@Tisoft:~/linux-command-practice/04-permissions$ umask
0022
tisoft@Tisoft:~/linux-command-practice/04-permissions$ umask -S
u=rwx,g=rx,o=rx
tisoft@Tisoft:~/linux-command-practice/04-permissions$ ls -ld private/
drwx----- 2 tisoft tisoft 4096 Sep 21 08:15 private/
tisoft@Tisoft:~/linux-command-practice/04-permissions$ ls -l private/secret.txt
-rw----- 1 tisoft tisoft 44 Sep 21 08:15 private/secret.txt
tisoft@Tisoft:~/linux-command-practice/04-permissions$ stat -c 'owner=%U group=%G permissions=%A' hello.sh
owner=tsift group=tsift permissions=-rwxr--r--
tisoft@Tisoft:~/linux-command-practice/04-permissions$ |

```

首次运行 `hello.sh` 提示权限不足，是因为文件没有执行权限；执行 `chmod u+x hello.sh` 后运行正常。

十四、系统信息命令（uname、uptime、w、free、df、du）

```

tisoft@Tisoft:~/linux-command-practice/04-permissions$ uname -a
Linux Tisoft 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026
x86_64 x86_64 x86_64 GNU/Linux
tisoft@Tisoft:~/linux-command-practice/04-permissions$ uptime
 08:51:46 up 48 min,  1 user,  load average: 0.00, 0.00, 0.07
tisoft@Tisoft:~/linux-command-practice/04-permissions$ w
 08:51:47 up 48 min,  1 user,  load average: 0.00, 0.00, 0.07
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
tisoft    pts/1    -                08:00   51:26   0.05s  0.04s  -bash
tisoft@Tisoft:~/linux-command-practice/04-permissions$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.7Gi        564Mi        6.8Gi        3.5Mi        449Mi        7.1Gi
Swap:          2.0Gi          0B          2.0Gi
tisoft@Tisoft:~/linux-command-practice/04-permissions$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G  0 3.9G  0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.9G  4.0K 3.9G  1% /mnt/wsl
drivers         200G  110G  90G  56% /usr/lib/wsl/drivers
/dev/sdd        1007G  1.6G  955G  1% /
none            3.9G  40K 3.9G  1% /mnt/wslg
none            3.9G  0 3.9G  0% /usr/lib/wsl/lib
rootfs          3.9G  2.8M 3.9G  1% /init
none            3.9G  536K 3.9G  1% /run
none            3.9G  0 3.9G  0% /run/lock
none            3.9G  0 3.9G  0% /run/shm
none            3.9G  80K 3.9G  1% /mnt/wslg/versions.txt
none            3.9G  80K 3.9G  1% /mnt/wslg/doc
C:\             200G  110G  90G  56% /mnt/c
D:\             277G  38G  240G  14% /mnt/d
tmpfs           786M  20K  786M  1% /run/user/1000
tisoft@Tisoft:~/linux-command-practice/04-permissions$ du -sh ~linux-command-practice
du: cannot access '~linux-command-practice': No such file or directory
tisoft@Tisoft:~/linux-command-practice/04-permissions$ du -sh ~/linux-command-practice
332K    /home/tisoft/linux-command-practice

```

第一次执行 `du` 时把路径写成了 `~linux-command-practice`，缺少 `/`；改为 `~/linux-command-practice` 后运行正常。

十五、压缩与解压命令（tar、gzip、gunzip、zip、unzip）

```

tisoft@Tisoft:~/linux-command-practice$ rm -rf work/archive-demo
tisoft@Tisoft:~/linux-command-practice$ mkdir -p work/archive-demo
tisoft@Tisoft:~/linux-command-practice$ cp -r 06-archive/project work/archive-demo/
tisoft@Tisoft:~/linux-command-practice$ cd work/archive-demo
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ tar -cvf project.tar project
project/
project/README.md
project/src/
project/src/run.sh
project/config/
project/config/app.conf
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ tar -tf project.tar
project/
project/README.md
project/src/
project/src/run.sh
project/config/
project/config/app.conf
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ gzip -k project.tar
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ ls -lh project.tar project.tar.gz
-rw-r--r-- 1 tisoft tisoft 10K Sep 21 08:56 project.tar
-rw-r--r-- 1 tisoft tisoft 379 Sep 21 08:56 project.tar.gz
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ cp project.tar.gz project-copy.tar.gz
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ gunzip -k project-copy.tar.gz
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ zip -r project.zip project
  adding: project/ (stored 0%)
  adding: project/README.md (deflated 6%)
  adding: project/src/ (stored 0%)
  adding: project/src/run.sh (stored 0%)
  adding: project/config/ (stored 0%)
  adding: project/config/app.conf (stored 0%)
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ unzip -l project.zip
Archive:  project.zip
  Length      Date    Time    Name
-----
      0  2026-09-21  08:56    project/
     72  2026-09-21  08:56    project/README.md
      0  2026-09-21  08:56    project/src/
     52  2026-09-21  08:56    project/src/run.sh
      0  2026-09-21  08:56    project/config/
     29  2026-09-21  08:56    project/config/app.conf
-----
    153                      6 files
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ mkdir unzip-result
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ unzip project.zip -d unzip-result
Archive:  project.zip
   creating: unzip-result/project/
  inflating: unzip-result/project/README.md
   creating: unzip-result/project/src/
 extracting: unzip-result/project/src/run.sh
   creating: unzip-result/project/config/
 extracting: unzip-result/project/config/app.conf
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ find unzip-result -maxdepth 3 -type f
unzip-result/project/README.md
unzip-result/project/src/run.sh
unzip-result/project/config/app.conf
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ |

```

十六、进程管理命令 (ps、jobs、bg、fg、kill、nice、nohup)

```
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ sleep 300 &
[1] 2744
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ jobs
[1]+  Running                  sleep 300 &
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ ps -ef | grep '[s]leep 300'
tisoft      2744      374  0 08:57 pts/0    00:00:00 sleep 300
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ kill -STOP %1
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ jobs
[1]+  Stopped                  sleep 300
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ bg %1
[1]+ sleep 300 &
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ jobs
[1]+  Running                  sleep 300 &
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ fg %1
sleep 300
^Z
[1]+  Stopped                  sleep 300
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ bg %1
[1]+ sleep 300 &
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ jobs
[1]+  Running                  sleep 300 &
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ kill %1
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ jobs
[1]+  Terminated             sleep 300
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ nice -n 10 sleep 60 &
[1] 2747
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ nohup sleep 60 > ~/linux-command-practice
/work/nohup-demo.log 2>&1 &
[2] 2748
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ ps -o pid,ni,cmd -C sleep
  PID  NI  CMD
  2747  10  sleep 60
  2748   0  sleep 60
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ killall sleep
[1]-  Terminated             nice -n 10 sleep 60
[2]+  Terminated             nohup sleep 60 > ~/linux-command-practice/work/nohup-demo.log 2>&1
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ |
```

十七、网络命令 (ip、ping、ss、curl、wget)

```

tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:e5:68:b8 brd ff:ff:ff:ff:ff:ff
    inet 172.19.152.85/20 brd 172.19.159.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fee5:68b8/64 scope link
        valid_lft forever preferred_lft forever
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ ip route
default via 172.19.144.1 dev eth0 proto kernel
172.19.144.0/20 dev eth0 proto kernel scope link src 172.19.152.85
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.488 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.047 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.042 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.078 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3214ms
rtt min/avg/max/mdev = 0.042/0.163/0.488/0.187 ms
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ ss -tuln
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port      Process
udp        UNCONN     0           0            127.0.0.54:53           0.0.0.0:*
udp        UNCONN     0           0            127.0.0.53%lo:53       0.0.0.0:*
udp        UNCONN     0           0            10.255.255.254:53      0.0.0.0:*
udp        UNCONN     0           0            127.0.0.1:323         0.0.0.0:*
udp        UNCONN     0           0            [::1]:323             [::]:*
tcp        LISTEN     0          4096         127.0.0.53%lo:53       0.0.0.0:*
tcp        LISTEN     0          1000        10.255.255.254:53      0.0.0.0:*
tcp        LISTEN     0          4096         127.0.0.54:53         0.0.0.0:*
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ curl -I https://example.com
HTTP/2 200
date: Mon, 21 Sep 2026 01:00:12 GMT
content-type: text/html
server: cloudflare
last-modified: Tue, 15 Sep 2026 23:41:26 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 8
cf-cache-status: HIT
cf-ray: a3e521b09b230892-HKG
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ wget --spider https://example.com
Spider mode enabled. Check if remote file exists.
--2026-09-21 09:00:29-- https://example.com/
Resolving example.com (example.com)... 172.66.147.243, 104.20.23.154, 2606:4700:10::6814:179a, ...
Connecting to example.com (example.com)[172.66.147.243]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Remote file exists and could contain further links,
but recursion is disabled -- not retrieving.
tisoft@Tisoft:~/linux-command-practice/work/archive-demo$ |

```

十八、作业感受和建议

学习感受

- 通过本次实践，我熟悉了 Linux 常用命令的基本用法，对文件操作、内容搜索、权限管理、压缩解压、进程管理和网络查看有了更直观的认识
- linux的各种指令对于操作linux系统至关重要，尤其是在AI时代更需要这种能力

意见与建议

- 学习到的指令通常跟着各个参数和选项，在使用的过程中很多时候都需要现场查，需要勤加练习才能日常使用的时候记得住

- 建议课程增加综合练习